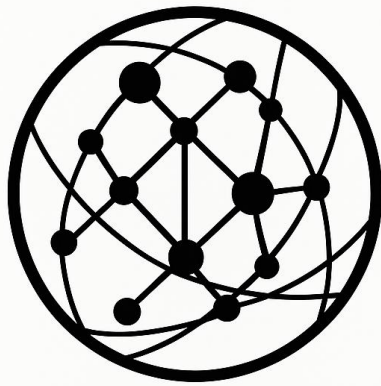


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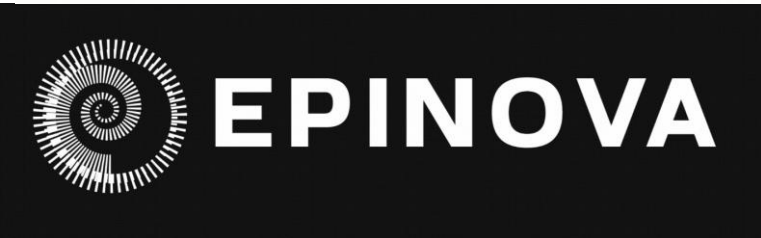
Series Information:
This policy brief is part of the EPINOVA Policy Brief Series on Strategic Competition, AI-Enabled Warfare, and Information Conflict.

Recommended Citation:
Wu, Shaoyuan (2026), *From Transit Fees to a Chokepoint-Hydrocarbon Value System: Repricing Passage, Energy Dependence, and Regional Leverage in the Strait of Hormuz*, Policy Brief No. EPINOVA-2026-PB-48, Global AI Governance and Policy Research Center, EPINOVA LLC.
<https://doi.org/10.67037/epinova.pb.2026.048>.

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From Transit Fees to a Chokepoint-Hydrocarbon Value System:
Repricing Passage, Energy Dependence, and Regional Leverage in the Strait of Hormuz

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Date: May 13, 2026

Key Judgments

- **Iran may be attempting to convert Hormuz passage uncertainty into a chokepoint-hydrocarbon value system.** The reported toll mechanism should be understood not only as a revenue tool, but as a broader effort to link maritime access, hydrocarbon dependency, security risk, alternative settlement channels, and political alignment.
- **The system remains embryonic and contested.** Public evidence does not yet confirm a stable, formal, state-recognized toll regime. Nor does it confirm that major oil and gas companies or leading shipping firms have openly accepted Iranian passage fees.
- **The strategic significance lies in repricing passage, not only collecting fees.** Even limited or deniable payment arrangements could create precedent if they are repeated, normalized, or embedded in commercial risk calculations.
- **The enforcement problem is shifting from maritime tracking to financial and logistical verification.** AIS data can show whether vessels move, but it cannot reveal whether passage was priced, cleared, insured, politically negotiated, or financially intermediated.
- **A partial CHVS would complicate sanctions enforcement and regional maritime governance.** If passage conditions are linked to alternative currencies, crypto assets, barter, offsets, informal swaps, or third-party intermediaries, the system could weaken visibility inside dollar-centered enforcement mechanisms.

Executive Summary

The reported emergence of a Hormuz transit-fee mechanism should not be understood only as a maritime toll dispute. Its broader significance lies in Iran’s possible attempt to convert chokepoint exposure, hydrocarbon dependency, maritime-risk pricing, and alternative settlement channels into a new form of regional value capture.

This brief proposes the concept of a **Chokepoint-Hydrocarbon Value System**, or CHVS, to describe this emerging logic. Under this model, the Strait of Hormuz is not simply a geographic passageway or military chokepoint. It becomes a contested governance space in which passage rights, security assurances, energy flows, payment channels, sanctions exposure, and political alignment may become increasingly linked.

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The evidence remains incomplete. Public reporting has not yet established a verified list of states, oil and gas companies, shipping firms, charterers, traders, or cargo owners that have formally paid Iran a Hormuz transit fee. Some reports point to vessel-level or operator-level payment pressure, while other reporting includes explicit denials by major maritime actors. The CHVS should therefore not be described as a fully institutionalized system.

The more defensible assessment is that Iran appears to be testing a selective passage-pricing regime. Its strategic importance does not depend only on whether each payment can be publicly verified. The deeper issue is whether Iran is attempting to reprice passage itself and convert maritime uncertainty into regional economic leverage.

Why This Matters

The CHVS model reframes the Hormuz crisis in three important ways.

First, it shifts the analytical focus from closure to pricing. The key question is not only whether ships can pass. It is under what cost, risk, delay, insurance, settlement, and political conditions they pass.

Second, it shifts the enforcement challenge from naval interdiction to financial and logistical verification. Payments may not appear as ordinary toll invoices. They may be routed through charterers, traders, cargo owners, intermediaries, bank guarantees, digital assets, insurance arrangements, cargo-linked compensation, or offset structures.

Third, it links maritime security to regional political economy. If Iran can impose even partial passage pricing, it gains more than revenue. It gains a mechanism for sorting actors, rewarding alignment, raising sanctions friction, and asserting limited governance over a critical energy corridor.

This is why the toll issue should not be dismissed simply because verified payment evidence remains limited. The early strategic question is not whether Iran has already established a complete toll regime. It is whether Iran is trying to establish the principle that access through Hormuz can be repriced by Iranian power.

Methodological Note

This brief uses a qualitative analytical framework rather than a transaction-level payment dataset. It evaluates CHVS as an emerging mechanism based on public reporting on Hormuz toll pressure, OFAC sanctions guidance, shipping-company statements and denials, reported alternative settlement channels, maritime-risk behavior, and prior EPINOVA analysis of Hormuz friction management and threshold-delaying logistics. The model is intended to support monitoring and interpretation, not to verify specific payments.

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1. Conceptual Framing: From Toll Collection to Value Capture

The Hormuz toll issue is often framed as a question of whether Iran can charge ships for passage. That framing is too narrow.

A more useful approach is to ask whether Iran is attempting to convert maritime uncertainty into a broader system of value extraction. Under this logic, the toll is not only a fee. It is a signal that passage through Hormuz may no longer be treated as cost-neutral, politically neutral, or financially external to Iranian power.

This brief defines the resulting structure as a Chokepoint–Hydrocarbon Value System. The CHVS refers to an emerging mechanism through which a state located near a critical energy chokepoint seeks to link five variables: passage access, hydrocarbon dependence, security risk, payment channels, and political alignment.

This model does not imply that Iran can fully control Hormuz. Nor does it suggest that Tehran can replace the established international maritime order. Rather, it suggests that Iran may be trying to create a parallel, friction-based value system inside a contested maritime space.

The core analytical point is therefore not whether every reported payment can be verified. It is whether the rules of passage are being contested in a way that allows Iran to extract value from uncertainty itself.

2. The Logic of Chokepoint–Hydrocarbon Value Capture

Iran’s potential leverage rests on the interaction of five structural elements.

First, **geographic concentration gives Hormuz enduring strategic value**. Even when traffic is degraded rather than fully blocked, the strait remains central to oil, gas, refined-product, LNG, LPG, shipping, and maritime-insurance calculations. A partially disrupted chokepoint can still generate systemic costs.

Second, **hydrocarbon dependency turns passage uncertainty into economic exposure**. Energy importers, refiners, traders, charterers, insurers, and cargo owners all face higher risk when transit becomes conditional, delayed, or politically differentiated.

Third, **security risk can become monetizable**. If passage is no longer treated as automatic, then risk reduction, tacit clearance, escort, protection, non-interference, or preferential handling may acquire economic value.

Fourth, **alternative settlement channels may allow value extraction outside conventional dollar-centered systems**. Reported references to yuan settlement, crypto payments, offsets, swaps, barter, or in-kind arrangements suggest that the toll issue may also function as a test of sanctions-circumvention and financial-sovereignty mechanisms.

Fifth, **political alignment may become embedded in logistics**. If “friendly” actors receive smoother passage, lower exposure, or preferential clearance, maritime access becomes part of a wider regional bargaining system. In that environment, logistics does not merely move goods. It sorts political relationships.

Together, these dynamics suggest a shift from simple coercion to structured value capture. Iran would not need to close the strait to generate leverage. It would only need to make passage more conditional, costly, selective, and politically legible.

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3. The Five-Layer CHVS Model

The Chokepoint–Hydrocarbon Value System can be understood as a five-layer analytical structure linking geographic control, hydrocarbon dependency, passage pricing, alternative settlement channels, and regional-order formation.

This layered structure matters because the reported Hormuz toll issue is not only maritime. It is simultaneously geographic, energy-related, financial, political, and institutional. Geography provides the bargaining foundation; hydrocarbon dependency turns disruption into systemic exposure; passage pricing monetizes uncertainty; alternative settlement channels reduce enforcement visibility; and political differentiation converts logistics into hierarchy and influence.

The model should therefore be treated as an analytical framework, not as a claim that Iran has already built a mature regional order. Its value lies in identifying how partial, deniable, and selectively enforced practices could evolve into a more repeatable mechanism if they are normalized over time.

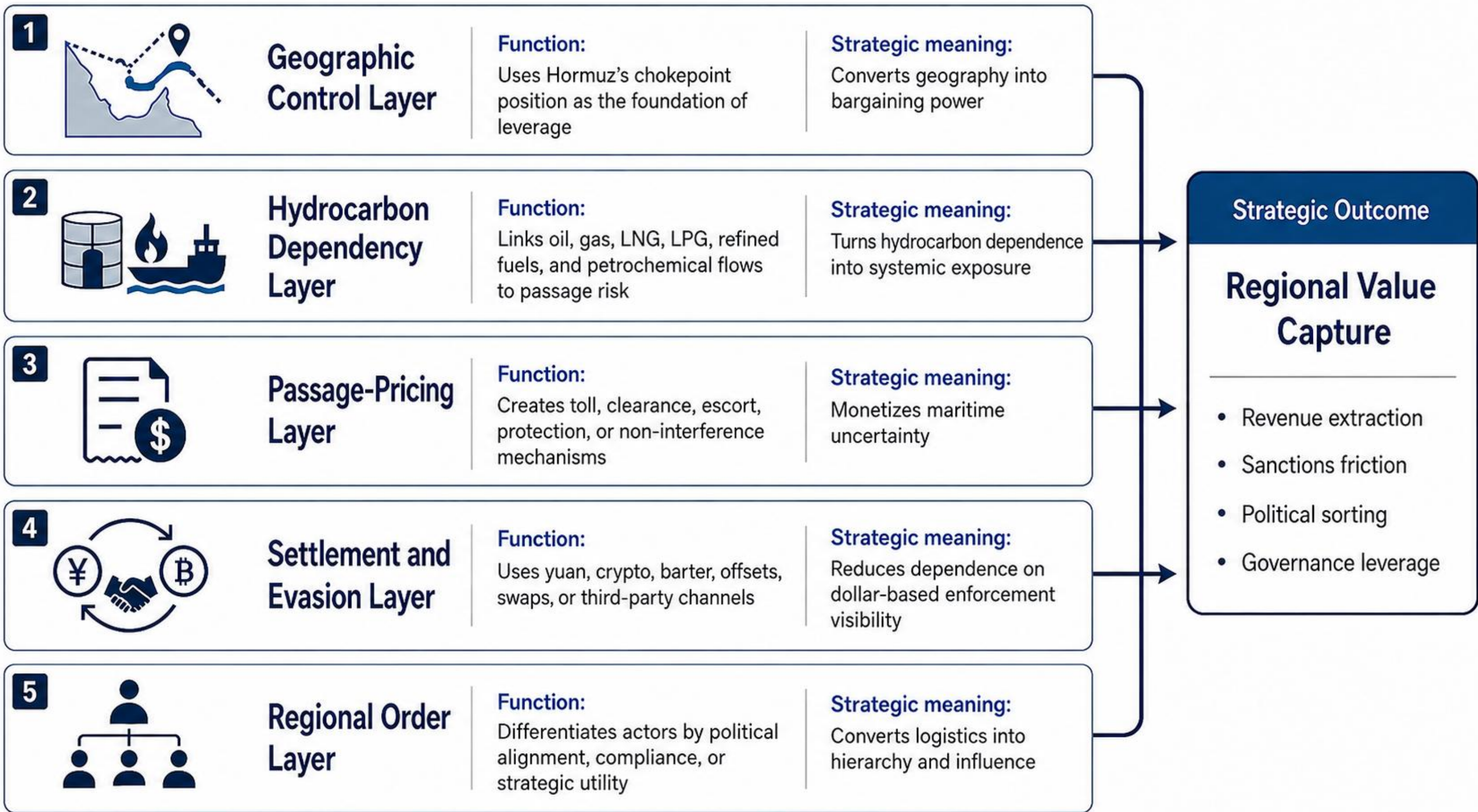


Figure 1. CHVS Model

Caption: The figure presents the Chokepoint–Hydrocarbon Value System as a five-layer model linking geographic control, hydrocarbon dependency, passage pricing, alternative settlement channels, and regional-order formation. It illustrates how maritime uncertainty around the Strait of Hormuz may be converted into regional value capture through revenue extraction, sanctions friction, political sorting, and governance leverage.

Source: Author’s analytical framework, building on EPINOVA’s prior analysis of Hormuz friction management, threshold-delaying logistics, and alternative corridor adaptation.

Note: CHVS is an analytical model describing an emerging mechanism, not a confirmed institutionalized system.

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3.1 Economic Logic of Value Capture

The CHVS model rests on a simple economic logic: Iran does not need to close Hormuz to extract value from it. It only needs to make passage sufficiently uncertain that clearance, non-interference, protection, or preferential treatment becomes economically meaningful.

In simplified terms:

$$VC_{CHVS} = T + RP + DC + SF + AP \quad (3.1)$$

where:

- **T** = direct toll or passage-related payment;
- **RP** = risk premium generated by insecurity or threat perception;
- **DC** = delay cost imposed by uncertainty, inspection, rerouting, or waiting time;
- **SF** = settlement-friction value generated through yuan, crypto, barter, swaps, offsets, or third-party channels;
- **AP** = alignment premium generated by preferential treatment for politically acceptable actors.

The model therefore treats formal tolls as only one component of value capture. Even without verified public payment records, Iran may still gain leverage if commercial actors adjust routing, insurance, settlement, or political behavior in response to perceived passage risk.

The toll is not the model. It is only the most visible component of a broader value-capture mechanism.

4. Evidence Boundary

The CHVS model should be presented as an emerging analytical framework, not as evidence of a completed toll regime.

Public reporting does not yet prove that a stable, formal, state-recognized Hormuz toll regime exists. Nor does it identify verified state, energy-company, shipping-company, trader, charterer, or cargo-owner payments. The strongest defensible claim is therefore narrower: Iran appears to be testing whether maritime passage uncertainty can be transformed into a value-capture mechanism linking energy flows, security risk, payment channels, and political alignment.

This distinction is essential. A confirmed toll regime would be an institutional fact. The current evidence points instead to an emerging coercive-pricing environment: partially visible, contested, deniable, and still under formation. For analytical purposes, CHVS should therefore be treated as a trajectory, not a completed system; its significance lies in direction, not confirmation.

5. Monitoring and Verification Challenges

The CHVS model suggests that monitoring should move beyond vessel movement alone. AIS data can show whether vessels enter or exit the strait, but it cannot show whether passage was priced, cleared, insured, politically negotiated, or financially intermediated.

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The most relevant indicators are therefore not only maritime. They include regulatory, financial, commercial, insurance, and legal signals.

Table 1. Key Monitoring Indicators for Hormuz Passage-Pricing and Value-Capture Dynamics

Monitoring Target	Why It Matters
OFAC designations and advisories	May reveal payment networks, intermediaries, sanctioned facilitators, or evasion channels
Company statements and denials	Help identify which firms resist, deny, or potentially comply with passage-pricing pressure
P&I club and marine-insurance notices	Show whether toll risk is being priced into coverage and liability structures
Charter-party disputes	May reveal who bears payment responsibility: owner, charterer, cargo owner, trader, or intermediary
Port-agent and customs documentation	Could expose clearance codes, guarantees, transit-linked payments, or documentation anomalies
Yuan, crypto, barter, or offset references	Indicate movement toward non-dollar settlement or sanctions-evasion mechanisms
Differentiated treatment of “friendly” vessels	Signals political sorting within the passage regime
Energy-trade routing shifts	Shows whether oil, gas, LNG, LPG, or refined-product flows are adapting to conditional passage
Insurance premiums and war-risk pricing	Indicates whether commercial markets are internalizing the passage-pricing regime

Source: Author’s analytical framework, informed by OFAC guidance on Hormuz toll-payment sanctions risk, public reporting on vessel-level toll pressure, shipping-company denials, alternative settlement channels, and Iran’s claimed initial toll revenue.

Note: Indicators are designed to identify whether passage through Hormuz is becoming priced, politically differentiated, financially intermediated, or embedded in broader sanctions-evasion and risk-allocation mechanisms.

The central policy challenge is therefore not only to prevent illicit payments, but to detect whether commercial actors are beginning to behave as though Iranian consent, clearance, or non-interference has acquired market value.

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6. Strategic Assessment

The CHVS remains embryonic. It is not yet a fully institutionalized regional order. But even in partial form, it could have important consequences.

A small number of payments, if repeated and normalized, could create precedent. A selective clearance system could reshape commercial expectations. A payment channel routed through non-dollar mechanisms could weaken sanctions visibility. A distinction between friendly and unfriendly vessels could convert maritime passage into an instrument of political alignment.

The system's value to Iran would not come only from revenue. It would also come from rule-setting. If Tehran can make commercial actors behave as though passage requires Iranian consent, then it has altered the operating assumptions of the strait even without imposing full closure.

This is the central strategic implication. The toll issue matters less as a confirmed fee schedule than as an early attempt to reprice passage itself.

7. Policy Implications

- **Treat Hormuz as more than a naval-access problem.** The crisis is also financial, insurance-related, legal, and political-economic. A vessel may pass physically while still being embedded in a coercive passage-pricing arrangement.
- **Monitor intermediaries, not only vessel owners.** Payment responsibility may fall on charterers, cargo owners, traders, brokers, insurers, port agents, or third-party financial facilitators. A narrow focus on shipowners may miss the actual payment chain.
- **Account for non-standard settlement.** If payments are routed through yuan, crypto assets, barter, offsets, swaps, in-kind compensation, or charitable fronts, ordinary dollar-clearing visibility may be reduced.
- **Incorporate passage pricing into energy-security analysis.** Oil and gas flows should be assessed not only by volume, but also by cost, delay, risk allocation, insurance exposure, and political dependence.
- **Distinguish tactical leakage from strategic normalization.** A few paid passages may not appear systemically important in volume terms. But if repeated, they could normalize the idea that Iran can price access through the strait.

8. Limitations

This brief evaluates an emerging mechanism, not a fully verified institutional system. Public reporting has not confirmed a formal Hormuz toll regime or identified verified state, energy-company, shipping-company, trader, or charterer payments.

Payment channels, if they exist, may be indirect and difficult to observe, including intermediaries, cargo-linked compensation, yuan settlement, digital assets, offsets, swaps, barter, or in-kind arrangements. Vessel movement through Hormuz does not prove payment.

The CHVS model is therefore analytical rather than conclusive. It identifies a plausible strategic logic linking toll pressure, energy dependence, maritime risk, alternative settlement, and political alignment. Future sanctions actions, company disclosures, insurance disputes, leaked documents, or official statements may strengthen or weaken this interpretation.

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Conclusion

The reported Hormuz transit-fee mechanism should be understood as more than a maritime toll dispute. It may represent an early attempt by Iran to transform chokepoint exposure into a broader system of regional value capture.

The Chokepoint–Hydrocarbon Value System remains incomplete, contested, and only partially visible. Public evidence does not yet confirm a formalized toll regime or provide a verified list of state, corporate, or energy-sector payers. But the absence of full confirmation does not eliminate the strategic significance of the attempt.

If Iran can establish even a partial expectation that passage through Hormuz is conditional, priced, politically differentiated, or financially intermediated, then the strategic meaning of the strait changes. Hormuz would remain a maritime chokepoint, but it would also become a monetizable governance space.

The core issue is therefore not only whether Iran has collected fees. It is whether Iran is trying to establish a new rule: that passage through Hormuz is no longer cost-neutral, politically neutral, or external to Iranian leverage.